



ISLEM REBHI

Engineering Student · Cloud & DevOps

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SUMMARY

Engineering student in Computer Science at ESPRIT, specializing in DevOps and Cloud computing, with experience in infrastructure deployment and fullstack development. Having worked on projects covering the entire application lifecycle.

EDUCATION

Engineering Degree in Cloud & DevOps **2023 – Present**

- ESPRIT – Ariana, Tunisia

Bachelor's Degree in Computer Science **2021 – 2023**

- ISITCOM – Hammam Sousse, Tunisia

High School Diploma in Mathematical Sciences **2020 – 2021**

- El Manar High School – Kasserine, Tunisia

TECHNICAL SKILLS

Cloud :

- **OpenStack** : Node Controller, Node Compute, Storage Node
- **Virtualization** : KVM / Hyper-V
- **IaaS / PaaS / SaaS**

DevOps :

- **Container** : Docker
- **Orchestration** : Kubernetes, Swarm
- **IAC et automatisaion** : Ansible, Terraform

Monitoring : Prometheus, Grafana, Zabbix, Nagios

System :

- **Linux** : Ubuntu, CentOS, RHEL
- **Administration** : SSH, Network management, Bash scripting, File sharing, Advanced storage management, User authentication, UNIX server security

Development and Scripting :

- **Backend** : Java, Spring Boot, Node.js, Flask, Symfony
- **Frontend** : Angular, React Native, HTML/CSS/JS, JavaFX
- **API** : REST
- **Scripting** : Python, Bash / Shell

Databases : PostgreSQL, MySQL, MongoDB, MariaDB

Software Architecture : Microservices, MVC

Machine Learning : Data preparation, supervised models (scikit-learn), analysis with pandas/NumPy

PROFESSIONAL EXPERIENCE

Full Stack Developer Intern

AL IDHAFA GROUP — Tunisia | July 2025 — August 2025

Participated in the development of Carnova, an intelligent platform dedicated to digitalizing the automotive market.

- Development of modular backend services in Spring Boot and REST API design.
- Creation of frontend components in Angular, integrating authentication and user role management (JWT).
- Modeling and optimization of PostgreSQL database.
- Implementation of a professional "Showroom" space for sellers.
- Stripe Checkout integration for secure payment.
- Addition of AI features: background removal (Rembg) & AI assistant (Gemma 1B via Ollama).
- Containerization of backend and frontend services via Docker.

Technologies : Spring Boot • Angular • PostgreSQL • Docker • Stripe • JWT • Ollama • Git

Mobile Developer Intern

ABAJIM — Sousse, Tunisia | February 2024 – June 2024

Participated in the development of a mobile application for visually impaired people, aimed at facilitating access to information through conversion of text content to audio.

- Design and development of a mobile application with React Native
- Integration of an OCR API for text recognition and extraction from documents (PDF, DOC) and images.
- Integration of a TTS (Text-to-Speech) API for converting extracted text into readable audio files.
- Implementation of an intuitive interface adapted to the needs of visually impaired users
- Backend development with Node.js for processing and data management
- Design of an intuitive interface adapted to the specific needs of the visually impaired.

Technologies : NodeJs • React Native • Mongo DB • Git

ACADEMIC PROJECTS

CityVoice – IaaS, Kubernetes Orchestration & Civic Platform

Academic — 2025-2026

Completion of a comprehensive project covering the entire application lifecycle, from infrastructure preparation (IaaS), automation, application development, to deployment and monitoring in an orchestrated containerized environment.

Phase 1 : Infrastructure Preparation (IaaS) :

- Deployment and configuration of main OpenStack services (Nova, Neutron, Glance, Keystone, Horizon, Cinder).
- Creation and management of virtual instances with complete network access (Internet, internal network).
- Implementation of a virtual network with routing, IP address allocation, and inter-instance communication management.
- Use of Heat to automate infrastructure deployment.
- Setup of reproducible infrastructure via templates.
- Deployment of a WAF (Web Application Firewall) with custom rules.
- Configuration of Security Groups and firewall rules to control network traffic.
- Implementation of resource access security mechanisms.
- Deployment of a monitoring tool (Zabbix) for infrastructure supervision, system resource monitoring (CPU, RAM, network), and anomaly detection and analysis.
- Implementation of a load balancer (Octavia) for load distribution.

Phase 2 : Automation & Monitoring :

- Automated deployment of a Kubernetes cluster (kubeadm) via Ansible.
- Structured into 7 professional roles: common, containerd, kubernetes, k8s-master, k8s-worker, kubectl-config, monitoring.
- Automation of dependency installation, control plane initialization, and worker node joining.
- Setup of reproducible, modular, and maintainable infrastructure.
- Cluster composed of 1 master and 2 worker nodes, with containerd as runtime and Flannel for networking, ensuring inter-pod and inter-node communication.
- Integration of Prometheus, Grafana, AlertManager, and Node Exporter.
- Collection and visualization of metrics: CPU, memory, network.

Phase 3 : Web Platform :

- Development of a web platform for citizen reporting of urban issues (roads, lighting, waste, etc.).
- Backend with Spring Boot: REST API design for managing reports, users, and interactions.
- Frontend with Angular (SPA): dynamic interfaces for submitting and viewing reports.
- Integration of geolocation enabling display and filtering of reports by area.
- Implementation of a report tracking system (creation, consultation, evolution).
- Data management with MySQL.

Automobile Prediction & Recommendation System (Machine Learning)

Academic — 2025-2026

- Collection, cleaning, and preparation of an automobile dataset (handling missing values, outliers, categorical variable encoding).
- Exploratory data analysis (EDA) and segmentation via clustering techniques.
- Training of supervised machine learning models for predicting price, fuel type, and horsepower.
- Design of a recommendation system to suggest the most suitable vehicles based on a given budget.
- Development and deployment of an interactive web application with Flask (backend) and HTML/CSS/JavaScript (frontend) to expose models and visualize results.

Technologies : Python • scikit-learn • pandas • Flask • HTML/CSS/JS

Sabeel – Agricultural Management and Supervision Platform

Academic — 2024-2025

- Design of a digital platform dedicated to remote supervision of agricultural activities
- Implementation of centralized management features for crops, equipment, and resources, with real-time monitoring.
- Integration of a module for recruiting agricultural workers and accessing specialized information and advice.
- Development of a complete web application with Symfony (user interfaces, supervision dashboard, account management).
- Development of a desktop application with JavaFX, adapted to environments with limited connectivity.
- Contribution to optimizing agricultural resources and improving communication between farmers and workers.

Technologies : Symfony • JavaFX • MySQL • Git

LANGUAGES

- Arabic — Native language
- French — Fluent
- English — Fluent